MILITARY SPECIFICATION

LIGHT, DIAGNOSTIC-EXAMINATION

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a portable hand-held light which provides long-wave ultraviolet light and daylight fluorescent illumination, with operator observation capability through an integral magnifying lens.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.

MILITARY

MIL-P-116	Preservation-packaging, Methods of.
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

FSC 6530

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C73.11 Plugs and Receptacles - General Purpose 125 volt, 15 amperes, 2 pole, 3 wire Grounding Type.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

3. REQUIREMENTS

3.1 Fire and casualty hazards.

3.1.1 The contractor shall submit proof, prior to delivery, that the items to be delivered under the contract conform to the applicable requirements of the Underwriters' Laboratories, Inc. Proof shall consist of one of the following:

A. UL label, provided the contractor agrees to have Underwriters' Laboratories, Inc. forward a copy of the UL report directly to the Contracting Officer, upon the latter's request.

B. Laboratory report from a qualified independent laboratory. If an independent laboratory is used, it must be adequately equipped and competent to determine compliance with UL standards. The qualification of such laboratory shall be determined by the Government which reserves the right to require a facilities report and to conduct a facilities survey. Any laboratory report submitted shall include actual test methods, data taken, and results of tests. Detailed findings of examinations necessary to determine compliance shall also be included. The test report shall also have a section that describes the test sample in such a way that the report can be used to insure identity of production items with the test sample. This section shall be supplemented with construction information such as materials used, spacings between electrical components, types of bushings, photographs, wiring diagrams, sketches, and parts list, all appropriately identified. Compliance with this paragraph does not excuse compliance with all other requirements of the contract and specifications.

3.2 Materials. Materials used in the fabrication of the light shall be as hereinafter specified. Materials not specified shall be selected by the Contractor and shall be of the highest quality and suitable in all respects for the intended purpose.

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3.3 Design and construction. The light shall be constructed in accordance with figure 1. The unit shall consist of the main housing, a magnifying lens, switches, ballast, fluorescent lamps, lampholders, reflector, and a handle with swivel bracket. The light shall selectively deliver either long-wave ultraviolet light or daylight fluorescent light. The unit shall also provide magnification of the area observed. The light shall be provided with a 3-conductor line cord, not less than 6 feet in length. The line cord shall be terminated in a 3-prong male plug conforming to American National Standards Institute (ANSI) Standard C73.11. The plug shall meet the Underwriters' Laboratories requirements for Hospital Grade. The lamps shall be arranged so that one of each type is on either side of the magnifying lens with the ultraviolet lamps at the outer edges. The weight of the light shall not exceed 3 pounds.

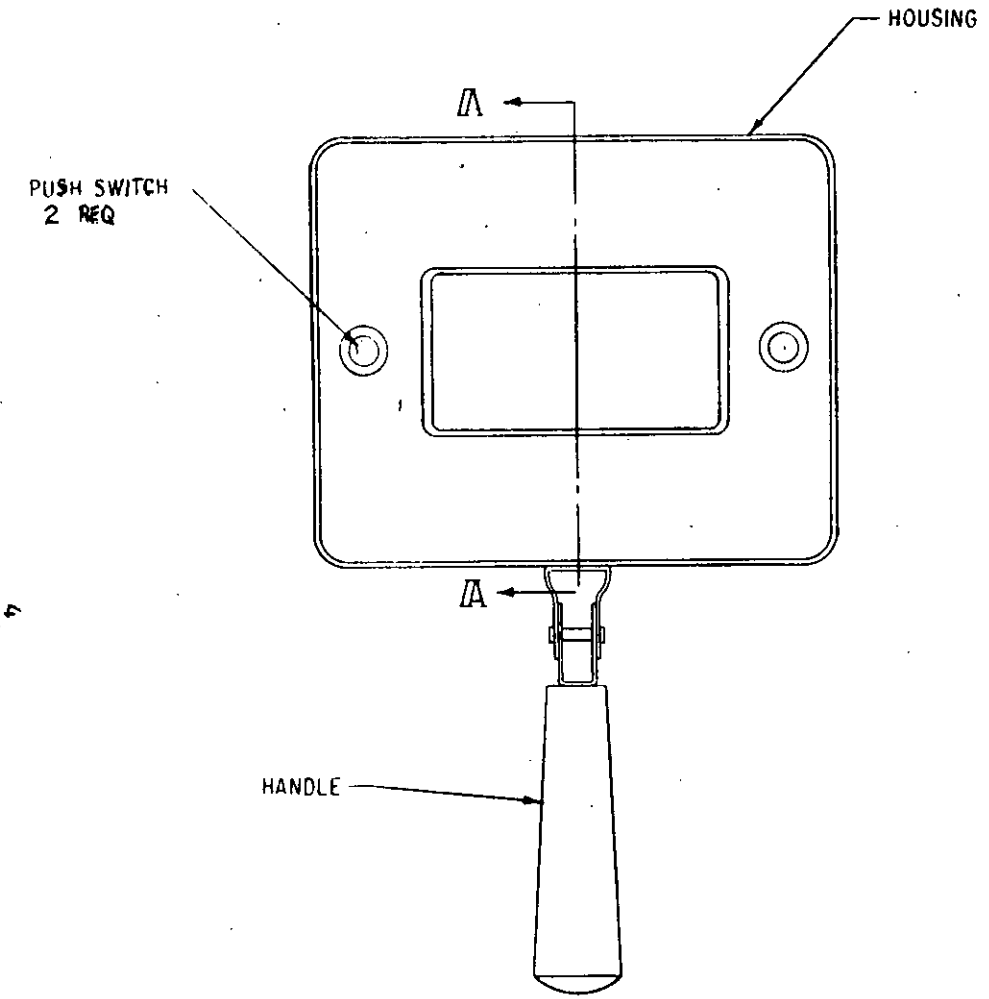
3.3.1 Housing. The housing shall be fabricated of molded phenolic, ABS plastic, or polycarbonate plastic. The housing dimensions shall not exceed 8 inches by 6 inches by 3 inches. The housing shall contain provisions for the mounting of the magnifying lens, reflector, switches, lampholders, ballast, and handle. The housing shall be rigid and shall be of sufficient thickness to preclude damage during normal usage.

3.3.2 Magnifying lens. The magnifying lens shall be optically ground and polished glass capable of providing a distortion free field of magnification. The focal length of the lens shall be approximately 8 inches. The magnifying lens shall be centrally mounted in the housing as shown in figure 1. The lens shall have nominal dimensions of 2 inches by 4 inches.

3.3.3 Switches. Two starter switches shall be provided, one securely mounted on either side of the magnifying lens as shown in figure 1. Where only one ballast is used, the switches shall be wired in such a manner as to allow only two lamps to be energized at one time. Where two ballasts are provided, one switch shall control one ballast.

3.3.4 Ballast. The unit shall be supplied with either one or two ballasts. The ballast shall be capable of providing power to two lamps at one time. Where one ballast is used, it shall be wired through the switches in such a way that actuating one switch shall energize daylight tubes, and actuating the other switch shall energize the two ultraviolet tubes. Where two ballasts are provided, one switch and one ballast shall control the ultraviolet lamps. The ballast(s) shall be mounted inside the housing behind the reflector. The ballast(s) shall be either secured to or in intimate contact with the reflector, in order to provide for heat dissipation. Starters used in conjunction with ballast shall not be acceptable.

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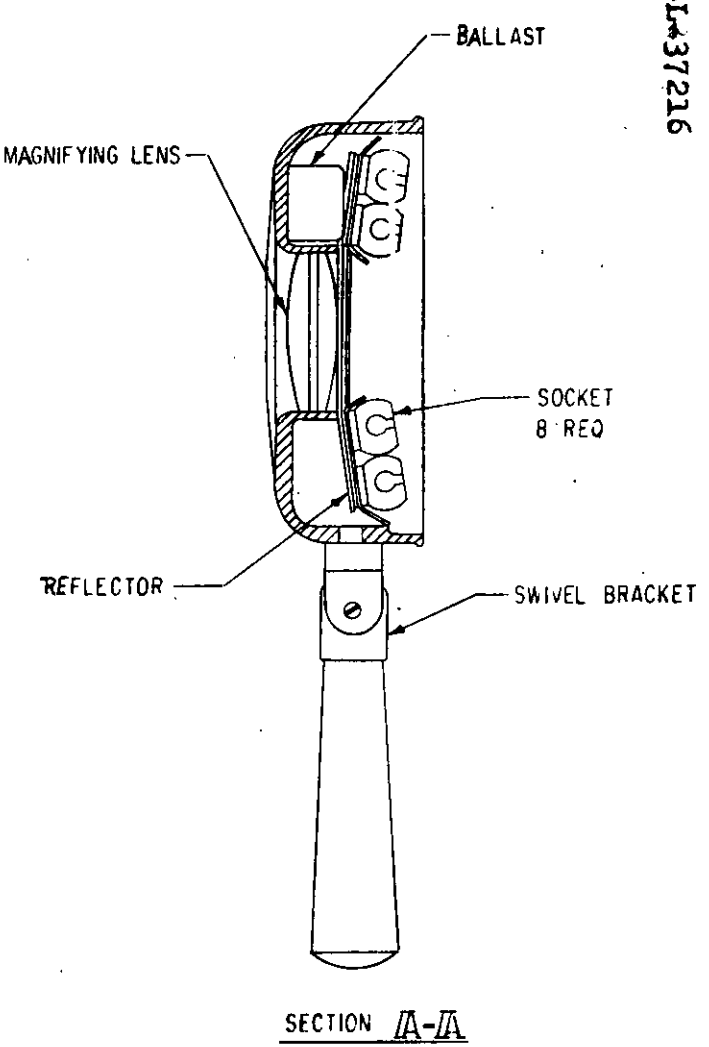


Figure 1. Light, Diagnostic-examination

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3.3.5 Fluorescent lamps. The unit shall be provided with four fluorescent lamps. Two lamps shall be ultraviolet fluorescent lamps, 6 inch, 4 watt, miniature bipin type (see 6.4). Two lamps shall be daylight fluorescent lamps, 6 inch, 4 watt, miniature bipin type (see 6.5). The ultraviolet lamps shall provide long wave ultraviolet light with a wavelength above 3200 Angstrom units and a peak average of 3500 to 3600 Angstrom units.

3.3.6 Lampholders. The light shall be equipped with eight miniature bipin lampholders suitable for use with the lamps specified in 3.3.5. The lampholders shall be securely mounted on the reflector in a manner to permit proper installation and removal of the lamps. The four lampholders on the outside shall be energized by the ultraviolet switch. The four lampholders closest to the magnifying lens shall be energized by the daylight switch.

3.3.7 Reflector. The reflector shall be fabricated of .0253 inch thick (No. 22 American Wire Gage) aluminum sheet or .0239 inch thick (No. 24 Manufacturer's Standard Gage) carbon steel sheet. Aluminum shall be anodized aluminum reflector sheet polished to a specular finish. Steel shall be chrome over nickel plated. The reflector shall be secured to the housing by means of screws to permit removal for repairs. The reflector shall be free of burs and sharp edges.

3.3.8 Handle. The handle shall be fabricated of molded phenolic, ABS plastic, or polycarbonate plastic. The handle shall be a minimum of 3.75 inches in length. The handle shall be either solid or hollow. The handle shall be secured to the housing by means of a chrome-plated or corrosion-resisting steel friction lock swivel joint.

3.3.9 Line Cord and plug. The light shall be provided with a 3-conductor line cord a minimum of 6 feet in length. A hospital grade 3-prong male plug shall be securely attached to the distal end of the line cord. The line cord shall be secured to the housing by means of a suitable strain relief. The line cord shall pass the pull test specified in 4.4.3 without displacement of the internal conductors. The line cord shall not be less than .0403 inch thick (No. 18 American Wire Gage). Line cord insulation shall be suitable for a minimum of 75 degrees Centigrade. The line cord shall enter the housing in the area of the swivel braket connection.

3.3.10 Electrical connections. The electrical connections shall be secure and shall be made so as to preclude contact with other conductors or dead metal parts. Wire nuts shall be used where applicable. Other connections shall be made in a manner acceptable to the Underwriters' Laboratories. All internal wiring shall be behind the reflector, protected from mechanical damage.

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3.4 Performance. The light shall operate on 120 volt 50/60 Hertz electric power. The unit shall be capable of operating continuously without creating a hazard due to overheating of components. Activating the proper switch shall cause the lamps controlled by it to light (see 4.4.1). The housing shall be capable of being easily adjusted at the swivel joint and shall firmly maintain any set position. The light shall be capable of withstanding the dielectric strength test specified in 4.4.2 without breakdown.

3.5 Workmanship. Workmanship shall be first class throughout. Light shall be free from defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual and auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the component or material, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the NSN, the Item Identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and table I.

TABLE I. Sampling for examination.

	Inspection level	AQL (defects per hundred units)
For visual examination		
Major defects	I	1.0
Minor defects	I	4.0
For dimensional examination	S-I	2.5

4.2.2 For tests. All units shall be subjected to the operation and dielectric tests. One unit shall be subjected to the strain relief test.

4.3 Examination. The lights shall be examined to determine compliance with all requirements contained in this specification.

4.3.1 Classification of defects. Examination shall be conducted in accordance with the following classification of defects, as shown in table II. Examination shall not be limited to the classified possible defects listed below.

TABLE II. Classification of Defects

Categories	Defects
Major	
101	Any component not free of breaks, cracks or sharp edges.
102	Light not complete with all specified components.
103	Lamps not as specified.
Minor	
201	Screws, hardware not tight.
202	Components not held firmly in place in unit.
203	Line cord not properly secured to unit.
204	Lamp housing not easily adjustable on swivel.
205	Lamp housing does not maintain set position on swivel.
206	Ultraviolet lamps not in outside sockets.

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4.4 Tests.

4.4.1 Operation test. The light shall be connected to a 115 volt, 60 Hz electric power source. The switch that controls the daylight lamps shall be activated and the light observed for proper operation. The switch that controls the ultraviolet lamps shall be activated and the light again observed for proper operation. The test shall be repeated with the light connected to a 115 volt, 50 Hz electric power source.

4.4.2 Dielectric test. The light shall have a slowly increasing AC potential of 0 to 1240 volts applied between live and dead metal parts. The voltage shall be maintained at maximum for one minute. There shall be no indication of dielectric failure.

4.4.3 Strain relief test. A direct pull of 35 pounds shall be applied to the line cord for one minute. The force shall be applied normal to the plane of entrance of the line cord into the housing. The internal connections of the line cord shall not be displaced.

4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of section 5. Defects shall be scored as specified in table III. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2, and the acceptable quality level shall be 4.0 defects per hundred units.

TABLE III. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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5. PACKAGING

5.1 Packaging. Packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A. Each diagnostic examination light shall be packaged in accordance with MIL-P-116, method IC-2 except that a preservative shall not be utilized.

5.1.2 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be level A, B or C as specified (see 6.2).

5.2.1 Level A. Six lights, packaged as specified in 5.1.1, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Each fiberboard box shall be waterproofed in accordance with paragraphs 30.4 of PPP-B-636.

5.2.2 Level B. Six lights, packaged as specified in 5.1.1, shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1, or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II as specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Interior packages shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contract shall be shown.

5.3.2 Exterior container. Exterior containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. The light is intended for use in the diagnosis and examination of dermatological and other medical disorders.

6.2 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packing and packaging (see 5.1 & 5.2).

6.3 This specification covers the following item appearing the Federal Supply Catalog:

National Stock Number	Item Identification
6530-00-480-8286	LIGHT, DIAGNOSTIC-EXAMINATION, 115 VOLT, 60 Hz, AC

6.4 Ultraviolet fluorescent lamps shall be in accordance with NSN 6240-00-319-4549 (Commercial lamp Code F4T5/BLB), or equal.

6.5 Daylight fluorescent lamps shall be in accordance with NSN 6240-00-796-6240 (Commercial lamp Code F4T5/D), or equal.

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency-DM

Project No. 6530-1344
CS

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

LIGHT, DIAGNOSTIC-EXAMINATION MIL-L-37216

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

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